



Research paper

Discerning potable water sources using Monte Carlo based simulation for health risk assessment in Omu-Aran, Nigeria



O.O. Elemile^a, E.M. Ibitogbe^{a,*}, K.S. Obayomi^b, P.A. Trzcinski^c, B.M. Falana^d, E.M. Okon^d, S.O. Olajide^a, D. Arije^d, B.E. Tokula^e, O.O. Ibitoye^a

^a Department of Civil Engineering, College of Engineering, Landmark University, P.M.B. 1001, Omu-Aran, Kwara State, Nigeria

^b Department of Chemical Engineering, Curtin University, CDT 250, 98009, Miri, Sarawak, Malaysia

^c School of Civil Engineering and Surveying, University of Southern Queensland, West Street, 4350, Queensland, Australia

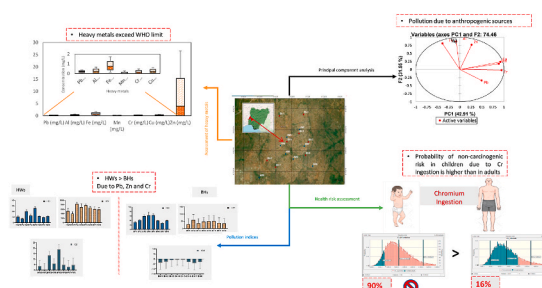
^d Department of Animal Science, College of Agricultural Sciences, Landmark University, P.M.B. 1001, Omu-Aran, Kwara State, Nigeria

^e Department of Environmental Management and Toxicology, Blue Gate Research Institute, Ibadan, Oyo State, Nigeria

HIGHLIGHTS

- The probability of exceeding HI value of 1 is greater in children (90%) than in adults (16%) upon exposure to Cr in HWs.
- Pollution indices reveal that BHs were of better quality while HWs were more polluted attributed to Pb, Zn and Cr.
- Source apportionment show presence of HMs are of anthropogenic and natural origin.
- Overall, BHs represent a safe water source for both adults and children, while HWs could pose health risk in children due to elevated HMs levels.
- Children are the most vulnerable receptors to carcinogenic and non-carcinogenic risk via ingestion.

GRAPHICAL ABSTRACT



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ABSTRACT

Human health is vulnerable upon exposure to persistent heavy metals (HMs) in environmental media such as groundwater even at trace levels. Therefore, this study aims to estimate the health risk posed by exposure to HMs using deterministic and probabilistic methods. Water samples were collected from 8 Hand dug wells (HWs) and 8 Boreholes (BHs) within Omu-Aran, Nigeria. The order of magnitude of the average contents for heavy metals are $Zn > Fe > Al > Cu > Cr > Mn > Pb$ in HWs and $Zn > Fe > Cu > Al > Cr > Pb > Mn$ for BHs. Pollution indices reveal that BHs were of better quality while HWs were more polluted. PCA shows contamination of HWs come from anthropogenic sources. Monte Carlo Simulation (MCS) technique was used to evaluate non-carcinogenic and carcinogenic risk in children and adults. The computed Hazard Index (HI) values for children and adults show the likelihood of exceeding the HI value of 1 is greater in children (90%) than in adults (16%) upon exposure to Cr in HWs. Computed lifetime carcinogenic risk (LCR) values via oral pathways for Pb exposure are well below the 10^{-4} , hence, no carcinogenic risk is expected upon exposure to Pb. Conversely, for HWs, LCR values for Cr exposure in children ranged from 0 to 2.14×10^{-4} , indicating that the current Cr levels pose a risk,

* Corresponding author.

E-mail address: ibitogbe.enoch@lmu.edu.ng (E.M. Ibitogbe).

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especially for children. Hence, prolonged duration to HMs exposure via ingestion increases the possibility of having cancer. Sensitivity analysis reveal Exposure frequency and HMs concentration are the most significant input factors on the risk estimate. Overall, BHs represent a safe water source for both adults and children, while HWs could pose health risk in children due to elevated HMs levels.

1. Introduction

Groundwater and surface water exist as natural water bodies and are the major sources of water globally. Groundwater is regarded as a much “cleaner” source and “free” from pollution when compared to surface waters due to infiltration and percolation (Arya et al., 2012; Boateng et al., 2016). Groundwater accounts for about 20% of water resources utilized for potable water supply and domestic uses (Pazand et al., 2018). Most countries are dependent on their groundwater resource as it is essential for domestic and industrial use as well as agricultural purposes etc., (Carrard et al., 2019; Wang et al., 2017). Concerns and uncertainties surround the sustainable exploitation of groundwater resources especially now that urbanization, population growth and industrialisation have been shown to have a direct impact on the use of this natural resource (Elemile et al., 2022; Iwuozor et al., 2022). Groundwater quality is just as critical as quantity for meeting water needs, which has prompted numerous studies worldwide to examine the quality of water in areas experiencing water scarcity (Tamasi, and Cini, 2004; Mohammadpour et al., 2022b). Human activities also have an effect on the quality of groundwater through direct or indirect pollution (Elemile et al., 2021). Fertilizer, effluent runoff from industries, open dump sites, and domestic sewage all contribute to groundwater pollution, resulting in health problems for those who consume them (Elemile et al., 2019). Additionally, Adeyemi and Ojekunle (2021) concludes that a serious global problem has developed as a result of various unsanitary activities that have resulted in heavy metals contaminating and polluting our groundwater. Heavy metals in various sources of drinking water are now widely recognized as toxic and a serious threat to human health, based on documented and recorded evidences from various part of the world. (Lu et al., 2018; Sankhla et al., 2016). Water quality monitoring and risk assessments of groundwater are critical as global development activities continue. This will aid in the development of a better understanding of potential changes in the quality of groundwater in all areas that rely on groundwater for irrigation and drinking water (Varol and Davraz, 2016). Ganiyu et al. (2021), observed the presence of Pb, Cd, Fe, and Zn above guideline values while investigating selected heavy metals (HMs) and its associated risks in groundwater samples in southwest Nigeria. Another investigation of groundwater samples within the vicinity of an oil depot in Nigeria showed high concentrations of lead. They attributed the source of lead contamination as emanating from actions of human including as the “loading” and “offloading” of petroleum products, (Oyeleke and Okparaocha, 2016; Olawale et al., 2020). Certain symptoms can be seen in people who are exposed to high levels of these metals, depending on what kind and how much they are present in the body (Adepoju-Bello and Alabi, 2005). According to Shankar et al. (2014), groundwater contamination has a negative impact on nearly 14% of the world’s population, which is significant. This shows the significance of monitoring water quality prior to consumption in order to protect human health. Understanding the sources, fate, and potential health risks posed by heavy metals in groundwater sources would prove to be a crucial step in determining the extent to which they exist. Furthermore, risk assessment has been demonstrated to be an effective tool for assisting decision makers and law enforcers in managing sites. (BIESIADA, 2001). Risk assessment is conventionally done using a deterministic (point estimate) approach where single values of variables are used which is reported to provide a simplistic assessment of the impact of pollutants (Giri and Singh, 2015; Li and Zhang, 2010; Tadiboyina and Ptsrk, 2016). According to (Adepoju-Bello and Alabi, 2005); much attention is paid to the numerical value of risk while

ignoring the uncertainty factor. However, researchers have begun considering uncertainty which is inherent in risk assessment (Pirsaheb et al., 2021; Sharafi et al., 2019), which is why uncertainty must be accounted for in the risk assessment process. Monte Carlo simulation is a widely used technique in accounting for the uncertainties inherent in a wide variety of risk-related problems. Within the validity of the assessment model, probability distributions are produced which highlights the likelihood towards exposure and health risks. Thus, it provides the decision-maker with a range of possible outcomes for an uncertain event, as well as the probability that the outcome will occur for any implementation choice (Wang et al., 2011; Rajasekhar, 2018). As a result, researchers (Joodavi et al., 2021; Mohammadpour et al., 2022; Mohammadpour et al., 2022b) have utilized the Monte Carlo Simulation (MCS) technique to assess the potential for hazards and pollution risk in a variety of environments. The aim of this study is to investigate the level of heavy metals found in groundwater sources utilized within the study area while using deterministic and probabilistic methods to estimate the health risk posed by heavy metal exposure.

2. Methods

2.1. Study area

The study area, Omu-Aran is geographically located at latitude 8°08′00″N and longitude 5°06′00″E, and is 564m above sea level (Fig. 1). It currently serves as the administrative headquarters of the Irepodun Local Government Area. The community is positioned within the southern limit of the tropical Savannah zone of Nigeria. Over the course of the year, the temperature typically varies with a uniformly high-temperature range from 25 °C to 30 °C with a mean annual rainfall range of 1200 mm (Akpenpuun, 2013; Kayode et al., 2022). Climate conditions in the study area alternate between wet and dry, referred to as rainy and harmattan respectively (Elemile et al., 2021a, 2021b). The wet season is generally warm, and overcast while the harmattan season is cold, windy, and partly cloudy. The dry season begins in November and continues until March of the following year, while the wet season typically lasts seven months (April to October).

More than 90% of the study area is underlain by the basement complex rocks that date back to the Precambrian and Cambrian ages while the remaining parts is underlain by Cretaceous and Younger sediments. Some of the most common rocks found in the study area include Granite-gneiss, biotite-granite, and metasediments like quartzite and quartz-mica schist. Elemile et al. (2019) reported that Hand-dug shallow wells (HWs) were the primary source of water for residents. The vast majority of the HWs used by its users are either outdated or in need of repair due to their age, ineffectiveness, or both. Deep well/-borehole (BH) sources are also found in the study area, as well as streams and wetlands.

2.2. Sample collection

Ground water samples (n = 48) were collected from the study area among sixteen (16) locations including 8 Hand dug wells (HWs) and 8 Boreholes (BHs) representing the most frequently used groundwater source within Omu-Aran, Kwara State. Using a GPS, the latitude and longitude of each sampling point were recorded at random, and these sources were selected at random. Three rounds of sampling were completed. Polyethylene bottles were used to collect and store the samples. Once the polyethylene bottles had been rinsed with pure water,

they were re-rinsed 2–3 times with the sample water as described by (Elemile et al., 2019). The samples were transported to the laboratory and acidified with the analytical reagent HNO_3 (5 percent v/v), after which they were sealed and stored at a low temperature of 4 °C. This was followed by determination of metal ion concentration (Cr, Fe, Cu, Al, Mn, Zn, and Pb) using an atomic absorption spectrometer (model AA – 6800). The research team carried out sampling campaigns between the months of October 2019 and December 2019, as well as between the months of October 2020 and December 2020, making a total of two campaigns. This was because the state was under a lockdown due to the COVID-19 outbreak, which limited people's mobility and movement during that time period.

2.3. Quality control

Water samples were analyzed in triplicate (documented as mean value) to ensure analytical precision, and the results demonstrated the effectiveness and sensitivity of the equipment in this regard. At the time of any sampling procedure, multiparameter equipment was calibrated. The concentration of trace elements found in blanks were only 3% of those measured in water samples, which allowed us to use these reagents to calibrate our instruments. Standard detection thresholds were observed throughout the metal concentration measurements, and reagent blanks were also checked throughout the process.

2.4. Heavy metal indices

2.4.1. Heavy metal evaluation index (HEI)

Heavy metal evaluation index (HEI) provides the overall water quality with respect to HMs. HEI is the same method (Edet and Offiong,

2002). Samples were calculated by dividing the concentration of a particular heavy metal by the maximum permissible concentration. Water quality becomes worse when the metal concentration is higher than the maximum acceptable level ($C_{i \max}$). The formula is given (Equation (1)).

$$\text{HEI} = \sum_{i=1}^n \frac{C_i}{C_{i \max}} \quad (1)$$

Where; $C_{i \max}$ is maximum allowable level and C_i is the concentration of individual element.

2.4.2. Heavy metal pollution index (HPI)

Water quality can be rated according to the cumulative impact of individual heavy metals on water quality using the heavy metal pollution index (HPI). It was first proposed by Mohan et al. (1996) and is computed as shown in equation (2).

$$\text{HPI} = \frac{\sum_{i=1}^n W_i Q_i}{\sum_{i=1}^n W_i} \quad (2)$$

Where Q_i refers to sub-index; W_i , the unit weight of the i th parameter; n , the total number of parameters. Furthermore, the sub-index (Q_i) is calculated according to eq. (3):

$$Q_i = \sum_{i=1}^n \frac{(M_i - I_i)}{S_i - I_i} \times 100 \quad (3)$$

where the monitored, ideal and standard heavy metal of i th parameter values are " M_i , I_i , and S_i ", respectively.

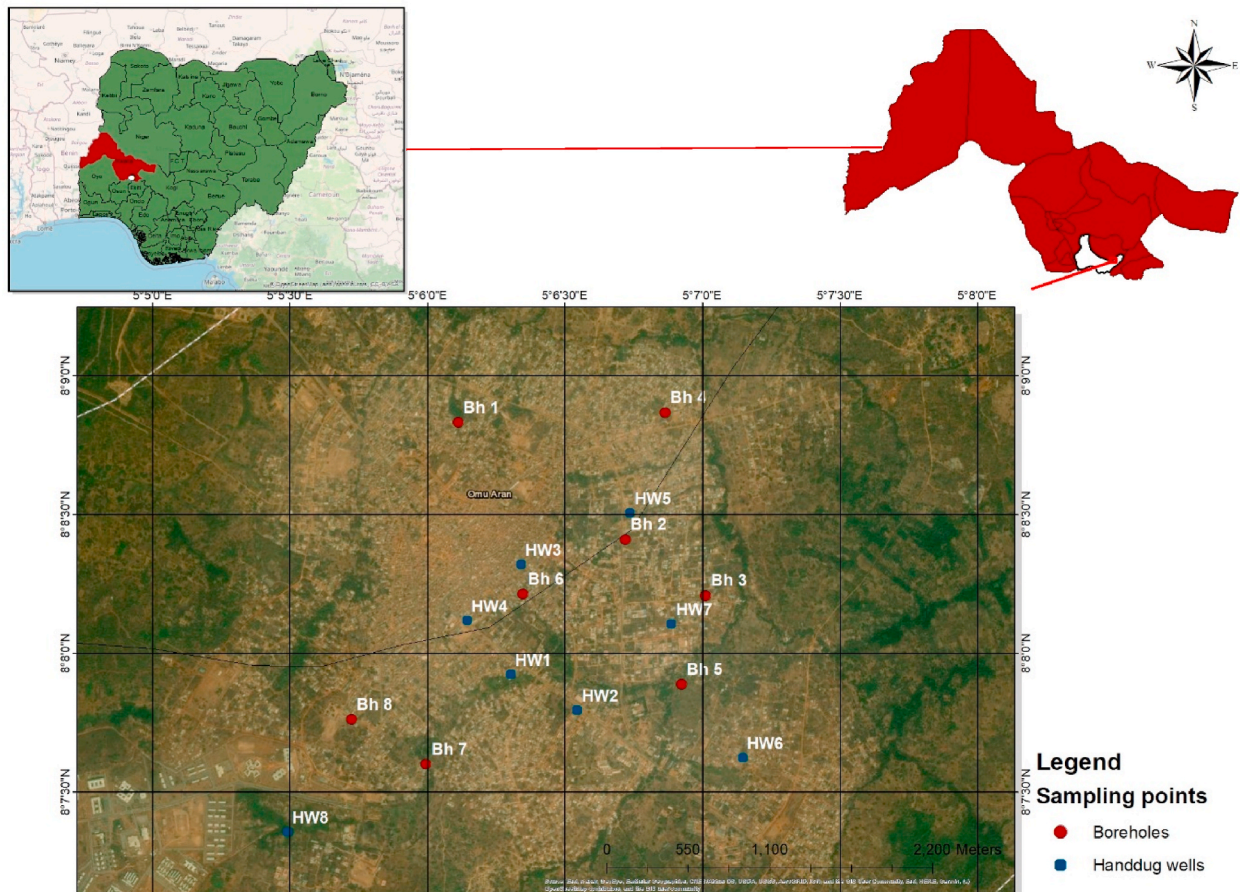


Fig. 1. Study Area showing sampling points for BHs and HWs.

2.4.3. Degree of contamination (C_d)

Degree of contamination index is used to aggregate the individual effects of several quality parameters deemed harmful on domestic waters (Backman et al., 1998).

$$C_d = \sum_{i=1}^n C_{fi} \quad (4)$$

$$C_{fi} = \frac{C_{Ai}}{C_{Ni}} - 1 \quad (5)$$

where the contamination factor, analytical value and upper permissible concentration are represented by C_{fi} , C_{Ai} and C_{Ni} , respectively.

In all cases, the maximum allowable level, $C_{i \max}$, standard values of the i th parameter, I_i and upper permissible concentration by C_{Ni} , used in this study are Pb (0.01 mg/L), Al (0.2 mg/L), Fe (3.0 mg/L), Mn (0.4 mg/L), Cr (0.05 mg/L), Cu (2.0 mg/L) and Zn (5.0 mg/L).

2.5. Health risk assessment

Human risk assessment is a critical technique for determining the probability of exposure to or risk of exposure to heavy metals over time (Wongsasuluk et al., 2014; Boateng et al., 2016). Generally, the first phase of risk assessment starts with identifying hazard, which is followed by exposure assessment, dose response and finally risk characterization.

The primary point of contact for heavy metals in groundwater are the dermal and oral pathways which according to Saha et al. (2017) are the two primary routes of exposure. Health risk assessment was carried out according to US EPA (2011) health risk assessment model for adult and infants.

2.5.1. Deterministic approach

Predicting the risk on human health due to exposure from heavy metals, chronic daily intake (CDI) ($\text{mgkg}^{-1}\text{day}^{-1}$) through incidental oral (CDI_{oral}) and dermal contact (CDI_{dermal}) was estimated from the resulting equations (Fang et al., 2014; US EPA, 2011). The parameters are displayed in Table 1. The equation is given as;

$$\text{CDI}_{\text{oral}} = \frac{(C_i \times IR \times CF \times FI \times EF \times ED)}{(BW \times AT)} \quad (6)$$

$$\text{CDI}_{\text{dermal}} = \frac{(C_i \times CF \times SA \times AF \times ABS \times EF \times ED)}{(BW \times AT)} \quad (7)$$

Hazard quotients (HQ), is defined as the ratio of the chronic daily intake (CDI) of individual metals to the oral reference dose (RfD) are used to compute non-carcinogenic risk for alternative oral and dermal pathways. Subsequently, the hazard index (HI), which indicates

cumulative non-carcinogenic effect is calculated as the sum of the respective hazard quotients (HQ). The respective RfD values (US EPA, 2016) for Cr, Mn, Cu, Zn, Pb, Al and Fe are 1.5×10^{-3} , 1.4×10^{-2} , 4.0×10^{-1} , 3×10^{-2} , 1.4×10^{-2} , 1×10^{-3} and 7×10^{-2} $\mu\text{g/kg/day}$ were used. A value of HI > 1 will mean adverse risk-effects that are non-carcinogenic in nature; a value less than 1 however represents no significant risk (USEPA, 2002). The HQ and HI are given in equations (3) and (4), respectively.

$$\text{Hazard quotient (HQ)} = \frac{\text{CDI}}{\text{RfD}} \quad (8)$$

$$\text{Hazard Index (HI)} = \sum \text{HQ} = \text{HQ}_{\text{oral}} + \text{HQ}_{\text{dermal}} \quad (9)$$

The lifetime cancer risk (LCR) was assessed when modeling the likelihood of carcinogenic risk using the IARC (1980) –“International Agency for Research on Cancer” framework. Heavy metals such as Cr and Pb are reported to have carcinogenic and neurological effect, respectively (Saleh et al., 2019; Balali-Mood et al., 2021). In order to obtain the LCR, the carcinogenic risk (CR) for each heavy metal is first calculated as the CDI multiplied by the cancer slope factor (SF) in (mg/kg/day). In this study, the SF considered Cr (5.0×10^{-1} mg/kg/day), and Pb (8.5×10^{-3} mg/kg/day), respectively. The calculations are shown in equations (5) and (6).

$$\text{Carcinogenic risk (CR)} = \text{CDI}_{\text{ingestion}} \times \text{SF} \quad (10)$$

$$\text{Lifetime cancer risk (LCR)} = \sum_i \text{CR}_i \quad (11)$$

According to US EPA (2016), the tolerable standard of LCR values ranges from 10^{-6} to 10^{-4} . The risk of cancer is negligible if LCR is less than 10^{-6} while any value greater than 10^{-4} is considered abnormal, and could lead to cancer over the course of a lifetime.

2.5.2. Probabilistic approach

Monte Carlo simulation is a computerized mathematical technique that rely on repeated random sampling to obtain numerical result which can be used to account for risk in quantitative analysis and decision making. It has been used successfully for representing quantitative events of uncertainties to obtain numerical outcome with the same level of precision. In practice, an approach that can handle numbers of uncertainties for numerous data in a risk assessment model is necessary (Qiu et al., 2021; Sadeghi et al., 2010). The risk calculation was repeated 10,000 times across different random values for all inputs (Table S3) during the Monte Carlo simulation process, and the output risk contains a range of values rather than a single value. There are outliers that can cause model values to diverge significantly between the “5th percentile” and the “95th percentile” when performing probabilistic risk calculations, so these two points were recommended as low and high estimates, respectively (Saha et al., 2017).

2.6. Data analysis

Data analysis involving Principal component analysis (PCA) was adopted to evaluate pollution sources using XLStat package. Probability risk analysis via Monte Carlo simulation was carried out to determine the risk estimation uncertainty with the aid of (Oracle Crystal Ball™). Arc Gis was used for mapping the position of the hand dug wells (HWs) and boreholes (BHs) investigated in the study area (see Fig. 1).

3. Result and discussion

3.1. Assessment of heavy metals

The concentration of seven HMs including lead (Pb), aluminum (Al), iron (Fe), manganese (Mn), chromium (Cr), copper (Cu), and zinc (Zn),

Table 1

Input parameters value for deterministic health risk assessment.

EXPOSURE PARAMETERS, SYMBOLS	UNIT	VALUE	REFERENCES
Concentration, C_i		from analysis	This study
Ingestion rate, IR	liters/day	2	USEPA (2000)
Conversion factor, CF	liter/cm ³	1×10^{-3}	United States (1989)
Fraction ingested, FI		1	–
Exposure frequency, EF	days/year	365	Mirzabeygi et al. (2017)
Exposure duration, ED	Years	70	(US EPA, 2011)
Body weight, BW	Kg	70	(US EPA, 2011)
Average time, AT	days/years	70×365	(USEPA, 2000)
Surface area, SA	cm ²	5700	(US EPA, 2011)
Adherence factor, AF		0.07	(Elemile et al., 2021)
Adsorption factor, ABS		0.001	(Elemile et al., 2021)

for both HWs and BHs were investigated in the study area. The results have been summarized and presented in Figs. 2 and 3, respectively. The average values of the HMs for HWs are Pb (0.079 ± 0.017 mg/L), Al (0.146 ± 0.099 mg/L), Fe (0.467 ± 0.365 mg/L), Mn (0.125 ± 0.234 mg/L), Cr (0.153 ± 0.175 mg/L), Cu (0.763 ± 1.738 mg/L), Zn (5.739 ± 6.01 mg/L). The results can be found in the supplementary file (Tables S1 and S2). The HMs concentration follow the decreasing order of magnitude as $Zn > Fe > Al > Cu > Cr > Mn > Pb$. Furthermore, it was observed that all elements investigated exceeded WHO (2017) safety limits. The average values of the heavy metals for BHs are Pb (0.004 ± 0.001 mg/L), Al (0.150 ± 0.022 mg/L), Fe (0.483 ± 0.182 mg/L), Mn (0.001 ± 0.000 mg/L), Cr (0.115 ± 0.029 mg/L), Cu (0.232 ± 0.088 mg/L), Zn (4.431 ± 5.406 mg/L). The HMs concentration follow the decreasing order of magnitude as $Zn > Fe > Cu > Al > Cr > Pb > Mn$ for BHs. Findings reveal that all HMs investigated within the study area for BHs exceeded WHO (2017) safety limits except for Pb, Al, Mn and Cu (see Tables S1 and S2).

3.2. Pollution indices

The use of pollution indices and their suitability for assessing fitness for human consumption has been highlighted as a critical tool for determining the collective impact of various heavy metals on overall water quality (Balakrishnan and Ramu 2016; Rizwan et al. 2011). In this study, three pollution indices namely heavy metal pollution indices, HPI, heavy metal index, HEI and degree of contamination, C_d , were used to evaluate water quality. The results are presented in Table 2. Due to the lack of a recommended critical value for HEI, workers must use their own discretion when evaluating pollution levels using this metric (Singha et al., 2020). The multiple of mean approach (Prasanna et al., 2012) was used to classify groundwater pollution (as determined by the results of the research). The proposed HEI value is classified as low ($HEI < 13$), medium ($HEI 13$ to 20) and high ($HEI > 20$) for HWs. Also, HEI was classified as low ($HEI < 4$), medium ($HEI 4$ to 6) and high ($HEI > 6$) for BHs. HEI values ranged from 5.58 to 10.84 for HWs with a mean of 13.71 (Fig. 4). From the above, 75% of samples fall within the low category while the remaining fall within the high zone. For BHs, values ranged from 2.80 to 6.39 with a mean of 4.59. We observe 50% of the samples fall within the low category while 25% fall within the medium and high zone each. The HPI values ranged from 400.460 to 838.153 with a mean of 636.87 for HWs and from 29.044 to 59.371 with a mean of 43.61 for BHs samples. The result shows that all HWs samples exceeded the Prasad and Bose (2001) recommended critical limit of 100 while BHs fall within the non-critical limit for drinking water. The C_d values ranged from 0.212 to 18.893 with a mean value of 6.71 for HWs and -4.205 to -0.610 with a mean of -2.41 for BHs (Fig. 5). C_d is generally classified into three groups: “low ($C_d < 1$), medium ($C_d = 1-3$) and high ($C_d > 3$)”

(Edet and Offiong 2002; Backman et al., 1998). We observe most of the samples (75%) from HWs exhibit high contamination whereas samples from BHs only show low contamination. In general, water samples from HWs are more polluted, whereas water samples from BHs tend to be of higher quality. The high values obtained for the pollution indices in this study were primarily due to Pb, Zn and Cr (see Table S9).

3.3. Pollution source identification

Source identification for heavy metals was done using PCA. As shown in Tables 3 and 4, the first three principal components (PCs) with eigenvalues greater than one were retained (See supplementary file Figs. S1 and S2). In total, they accounted for up to 92.63 and 86.99% of the variance in the water quality datasets for HWs and BHs, respectively. According to Gradilla-Hernández et al. (2020), acceptable values fall within 70% and 90%. For HWs, the first principal component (PC1) accounts for 42.91% of the total variability demonstrating a positive strong loading on Mn, Cr, and Cu. The positive strong loading of Mn and Cu supports the possible impact of natural processes. On the other hand, Cr is likely to come from anthropogenic sources probably because most HWs are improperly sealed leaving them exposed to the environment and influence of human activities on the surface. PC2 explains 31.55% of the total variance and has strong positive loadings on Al, Fe, and Zn. This would imply a more geogenic origin due to the predominant rock type. Biotite is rich in Al and quartzite has been enriched with iron oxide. In HWs, the result of the third principal component (PC3) is 18.17% of the total variance with positive strong and moderate loadings on Pb and Zn, respectively. Sources of Pb could be from anthropogenic sources. Moreover, the occurrence of Pb is due to leaded gasoline and exhaust from automobiles along the highway (Bhuiyan et al., 2010; Prasanna et al., 2012; Kiris and Baltas, 2021). For boreholes (BHs), PC1 accounts for 38.25% of the overall variation associated with Al, Cu, Fe and Zn. Zinc enrichment of groundwater could be connected to “water-rock interaction” such as the dissolution of Ba–Zn bearing rocks (Matiatos et al., 2014; Reza and Singh, 2019; Obayomi et al., 2021). Meanwhile, PC2 explains 27.93% of the total variance having positive loadings on Mn and negative loadings on Fe. The third principal component (PC3) explains 20.81% of the variance with strong loadings on Pb and Cr. The positive strong loading of Pb and moderate loading of Zn suggesting an interplay of anthropogenic and geogenic activities. As stated earlier, Zn enrichment is a result of “water-rock interaction” while it is likely that Pb is from anthropogenic origin. Pearson correlation showed that there is a very weak relationship between Pb and Zn which supports the fact that they do not originate from similar sources (as seen in supplementary file Table S9).

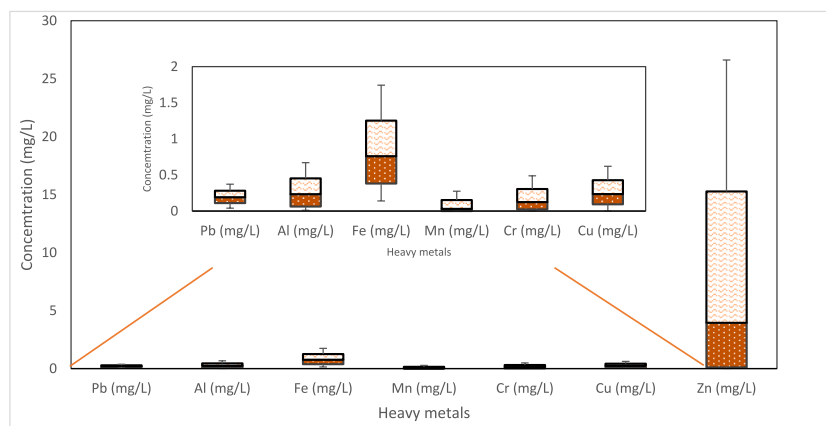


Fig. 2. Concentration of heavy metals in HWs.

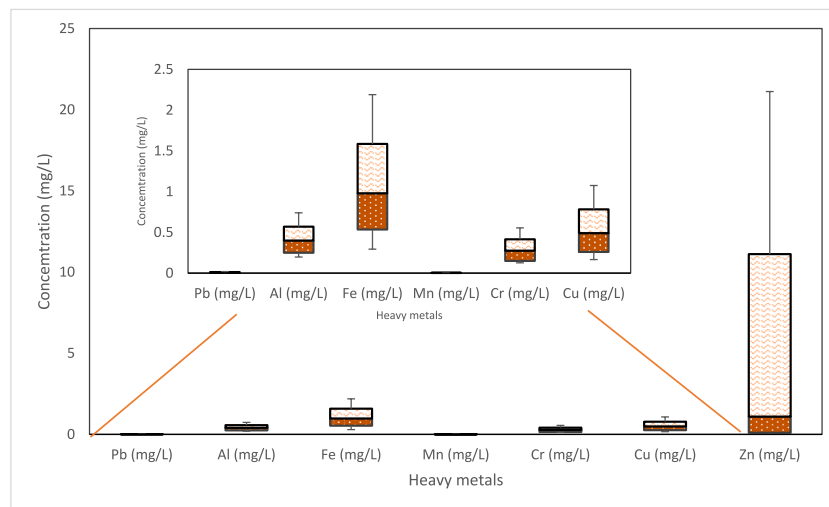


Fig. 3. Concentration of heavy metals in groundwater Concentration of heavy metals in BHs.

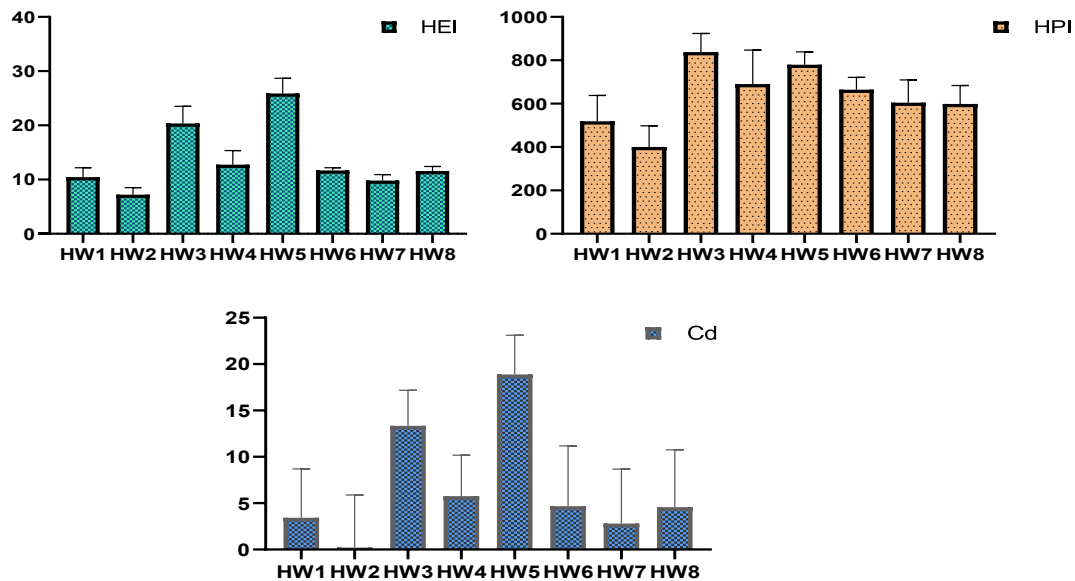


Fig. 4. Pollution Indices (a) HEI, (b) HPI and (c) C_d representing cumulative impact of various HMs in HWs.

Table 2
Pollution Indices and classification for Water sources.

Pollution Indices	Rating	Water sources	
		HWs	BHs
HEI	Low $_{HWs}$ (<13) Low $_{BHs}$ (<4)	HW1, HW2, HW4, HW6, HW7, HW8	BH1, BH6, BH7, BH8.
	Medium $_{HWs}$ (13–20) Medium $_{BHs}$ (4–6)	–	BH2, BH3
	High $_{HWs}$ (>20) High $_{BHs}$ (>6)	HW3, HW5	BH4, BH5
	Non-critical (<100)	–	All sources
HPI	Critical (≥ 100)	All sources	–
C_d	Low (<1)	HW2	All sources
	Medium (1–3)	HW7	–
	High (>3)	HW1, HW3, HW4, HW5, HW6, HW8	–

Low $_{HWs}$, Medium $_{HWs}$, High $_{HWs}$ – low, medium and high-level classification for HEI (HWs) and Low $_{BHs}$, Medium $_{BHs}$, High $_{BHs}$ – low, medium and high-level classification for HEI (BHs).

Table 3
Principal component score for HWs.

	PC1	PC2	PC3
Pb	0.494	−0.341	0.788
Al	−0.083	0.871	−0.201
Fe	−0.396	0.811	0.172
Mn	0.886	0.178	−0.409
Cr	0.947	−0.018	0.119
Cu	0.918	0.236	−0.264
Zn	0.267	0.767	0.575
Eigenvalue	3.00	2.21	1.27
Variability (%)	42.91	31.54	18.17
Cumulative %	42.91	74.46	92.63

3.4. Health risk assessment

3.4.1. Deterministic approach

Using the traditional deterministic assessment model, the HQ_{dermal} , hazard quotient by oral pathway (HQ_{ora}) and hazard index (HI) values were calculated for each location as summarized in Tables 5 and 6 for

Table 4
Principal component score for BHs.

	PC1	PC2	PC3
Pb	0.183	0.017	0.777
Al	0.856	0.494	0.106
Fe	0.569	−0.585	−0.444
Mn	−0.037	0.976	−0.069
Cr	0.058	−0.437	0.781
Cu	0.933	0.299	0.099
Zn	0.844	−0.368	−0.144
Eigenvalue	2.68	1.96	1.46
Variability (%)	38.251	27.929	20.81
Cumulative %	38.251	66.18	86.99

children and adults, respectively (HQ_{dermal} , HQ_{oral} and HI for each heavy metal is found in the supplementary file [Tables S4–S7](#)). It was observed that all hazard index values for all HWs with exception of HW5 were below 1. Since these values are below the stipulated [US EPA \(2011\)](#) recommendation, there will be no health hazards associated with use of shallow water sources (HWs) in the study area through any of the pathways (oral or dermal) for infants. In the case of BHs, all the computed HI values fall below one in line with the [US EPA \(2011\)](#) guideline indicating that there was no associated health risk attached to its use. Hence, users are not prone to health risk. Of course, health risk could occur for children as the result obtained for HW5 indicates that the HQ bears non-carcinogenic risk since it exceeds the recommended value. A similar trend was reported by Jiang *et al.*, (2021) who disclosed that $HI > 1$ indicates health risk to both adults and children. The reason for observed high levels at this location may be due to the position and condition of HW5 making it prone to contamination from elevated HMs levels. Because of its unpaved surroundings and proximity to busy roads,

we discovered during our investigation that the area was completely unprotected and constantly exposed to the environment. When non-carcinogenic risk was evaluated for the adult age group for both HWs and BHs, our findings clearly reveals that the water presents no hazard (non-carcinogenic) having no health implication for consumers since it falls within recommended value ($HI < 1$). In general, BHs are a source of water that is safe for both adults and children, whereas the high levels of HMs that are found in HWs could pose a potential health risk to children. The result for carcinogenic health risk via oral pathway for both children and adult is displayed in [Tables 7 and 8](#), respectively. In children the lifetime carcinogenic risk $LTCR_{child}$ values for Pb exposure ranged from 3.545×10^{-7} to 7.33×10^{-7} for HWs and 1.44×10^{-8} to 4.31×10^{-8} for BHs. Similarly, $LTCR_{adults}$ values for Pb exposure ranged from 7.59×10^{-8} to 1.57×10^{-7} for HWs and 3.08×10^{-9} to 9.24×10^{-9} for BHs. The values are well below the 10^{-4} , hence, no carcinogenic risk is expected upon exposure to Pb. The situation changes when Cr is considered. The $LTCR_{child}$ values for Cr exposure in children ranged from 0 to 2.14×10^{-4} for HWs and 3.24×10^{-5} to 6.2×10^{-5} for BHs. For adults, this value ranged from 0 to 3.0×10^{-5} for HWs and 6.49×10^{-6} to 1.33×10^{-5} for BHs. The target level of 1×10^{-4} was nearly reached for most of the values and surpassed for HWs. It is seen clearly that exposure to present Cr levels poses a threat especially amongst children. Hence, drinking (ingestion) of these water sources for a prolonged duration increases the possibility of having cancer.

3.4.2. Probabilistic approach

The Monte Carlo method using Oracle Crystal ball™ was adopted for simulating both non-carcinogenic (HI) and carcinogenic (LTCR) health risk value for adults and children under drinking water for HWs and BHs scenarios. Probabilistic estimates of Hazard Index (HI) for HWs and BHs at a 95% confidence level are displayed in [Figs. 6 and 7](#) respectively. A

Table 5
Non-carcinogenic health risk via dermal and oral pathways in children.

Heavy metals	Children															
	HW1		HW2		HW3		HW4		HW5		HW6		HW7		HW8	
	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ
	oral	dermal	oral	dermal	oral	dermal	oral	dermal	HQ oral	dermal	oral	dermal	oral	dermal	oral	dermal
HWs	0.239	0.001	0.225	0.003	0.313	0.036	0.266	0.002	1.234	0.054	0.135	0.013	0.104	0.012	0.123	0.013
HI = ΣHQs	0.24		0.228		0.349		0.268		*1.288		0.148		0.116		0.136	
BHs	BH1		BH2		BH3		BH4		BH5		BH6		BH7		BH8	
	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ oral	dermal	HQ	HQ	HQ	HQ	HQ	HQ
	oral	dermal	oral	dermal	oral	dermal	oral	dermal			oral	dermal	oral	dermal	oral	dermal
BHs	0.204	0.014	0.187	0.013	0.193	0.012	0.239	0.013	0.25	0.015	0.148	0.013	0.135	0.008	0.144	0.015
HI = ΣHQs	0.219		0.2		0.206		0.251		0.265		0.161		0.144		0.159	

* the HI value is greater than 1 representing risk (non-carcinogenic).

Table 6
Non-carcinogenic health risk via dermal and oral pathways in adults.

Heavy metals	Adult															
	HW1		HW2		HW3		HW4		HW5		HW6		HW7		HW8	
	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ
	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal
HWs	0.051	0	0.048	0.002	0.067	0.018	0.057	0.001	0.265	0.027	0.029	0.007	0.02	0.004	0.035	0.008
HI = ΣHQs	0.052		0.05		0.085		0.058		0.292		0.036		0.024		0.043	
BHs	BH1		BH2		BH3		BH4		BH5		BH6		BH7		BH8	
	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ	HQ
	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal	oral	dermal
BHs	0.04	0.004	0.04	0.007	0.041	0.006	0.051	0.006	0.054	0.007	0.032	0.007	0.029	0.004	0.031	0.008
HI = ΣHQs	0.044		0.047		0.048		0.058		0.061		0.038		0.033		0.038	

Table 7
Carcinogenic health risk via oral pathways in Children.

Heavy metals	Lifetime Cancer Risk							
	HW1	HW2	HW3	HW4	HW5	HW6	HW7	HW8
Pb	4.552E-07	3.545E-07	7.330E-07	6.324E-07	6.109E-07	6.324E-07	5.749E-07	5.606E-07
Cr	0.000E+00	9.864E-06	1.436E-04	5.989E-06	2.105E-04	5.355E-05	4.791E-05	5.073E-05
Heavy metals	Lifetime Cancer Risk							
	BH1	BH2	BH3	BH4	BH5	BH6	BH7	BH8
Pb	1.437E-08	4.312E-08	2.156E-08	2.875E-08	2.875E-08	3.593E-08	2.156E-08	1.437E-08
Cr	5.778E-05	5.355E-05	4.791E-05	5.073E-05	5.778E-05	5.355E-05	3.241E-05	6.200E-05

Table 8
Carcinogenic health risk via oral pathways in Adults.

Heavy metals	Lifetime Cancer Risk							
	HW1	HW2	HW3	HW4	HW5	HW6	HW7	HW8
Pb	9.753E-08	7.597E-08	1.571E-07	1.355E-07	1.309E-07	1.355E-07	1.232E-07	1.201E-07
Cr	0.000E+00	2.114E-06	3.077E-05	1.283E-06	4.5113E-05	1.15E-05	6.9451E-06	1.33E-05
Heavy metals	Lifetime Cancer Risk							
	BH1	BH2	BH3	BH4	BH5	BH6	BH7	BH8
Pb	3.08E-09	9.24E-09	4.62E-09	6.16E-09	6.16E-09	7.7E-09	4.62E-09	3.08E-09
Cr	6.94E-06	1.15E-05	1.03E-05	1.087E-05	1.24E-05	1.15E-05	6.9451E-06	1.33E-05

number of metals were investigated in the study, but detailed attention was given to Pb because of its toxicity and Cr since it is classified as a carcinogen. When values were simulated for non-carcinogenic risk, the values obtained at mean and 95th percentile levels showed risk was below the threshold value ($HI < 1$). However, upon exposure to chromium in the case of HWs, the respective mean and 95th percentile levels are 1.72 and 2.92 for children and 0.75 and 1.29 for adults indicating health risk (non-carcinogenic). The probability of exceeding this value is greater in children (~90%) than in adults (~16%). The HI from exposure to HMs for children and adults based on the Monte Carlo simulation are contained in the supplementary file as Fig. S3 (for HWs) and Fig. S4 (BHs).

Additionally, potential lifetime cancer risk (LTCR) for Pb and Cr were simulated for both water source types. For HWs, mean and 95th

percentile $LTCR_{child}$ values (7.26×10^{-8} , 1.4×10^{-7}) and $LTCR_{adult}$ values (4.16×10^{-8} , 7.41×10^{-8}) were observed for Pb. Likewise, mean and 95th percentile $LTCR_{child}$ values (2.1×10^{-5} , 4.57×10^{-5}) and $LTCR_{adult}$ values (1.2×10^{-5} , 2.46×10^{-5}) were observed for Cr. In the case of BHs, mean and 95th percentile $LTCR_{child}$ values (4.33×10^{-9} , 8.9×10^{-9}) and $LTCR_{adult}$ values (2.43×10^{-9} , 4.55×10^{-9}) were observed for Pb. Similarly, mean and 95th percentile $LTCR_{child}$ values (6.34×10^{-6} , 1.22×10^{-5}) and $LTCR_{adult}$ values (3.57×10^{-5} , 3.41×10^{-5}) were observed for Cr. The values obtained in this study is lower than the maximum acceptable carcinogenic risk recommended by the USEPA (1×10^{-4}). Furthermore, the likelihood of exceeding this value is almost zero.

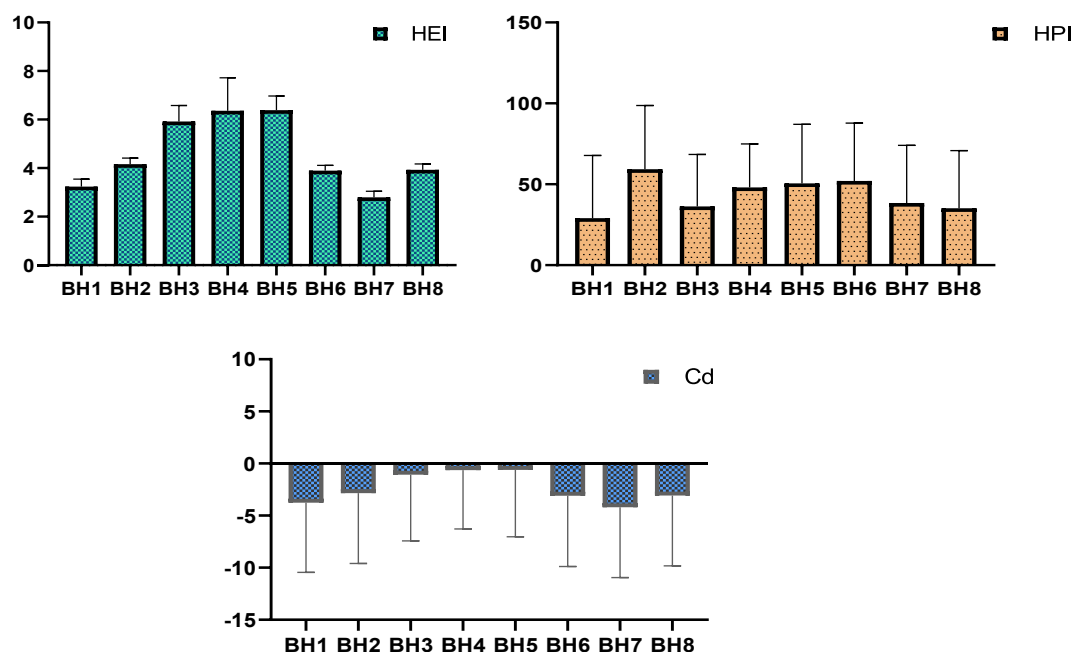


Fig. 5. Pollution Indices (a) HEI, (b) HPI and (c) C_d representing cumulative impact of various HMs in BHs.

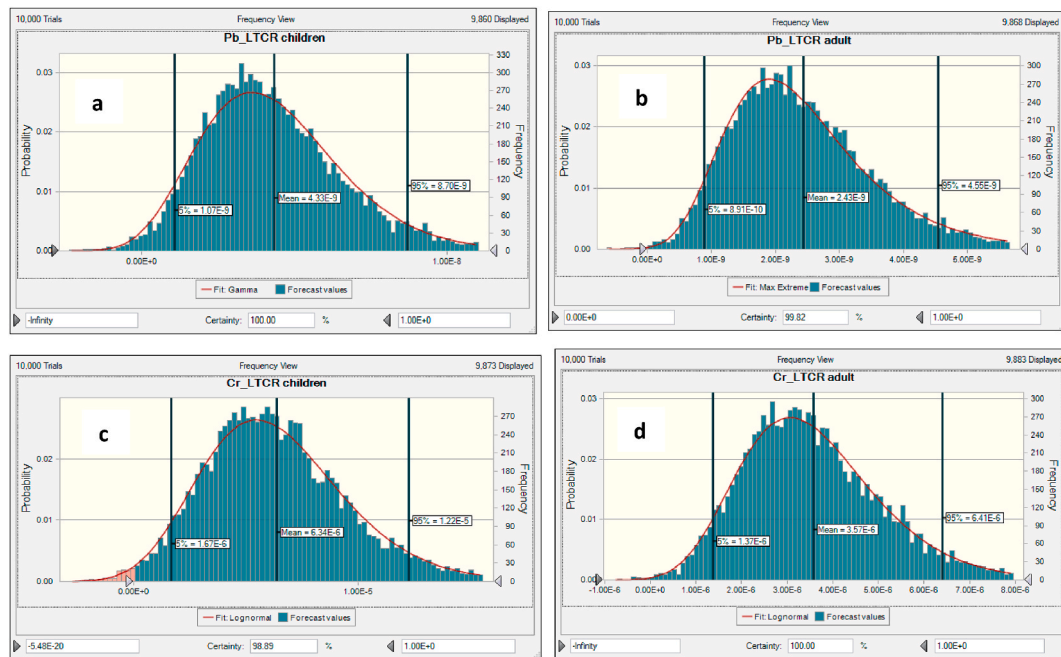


Fig. 6. Monte Carlo simulation for carcinogenic risk in HWs (a) exposure to Pb in case of children (b) and adults (c) exposure to Cr in case of children (b) and adults.

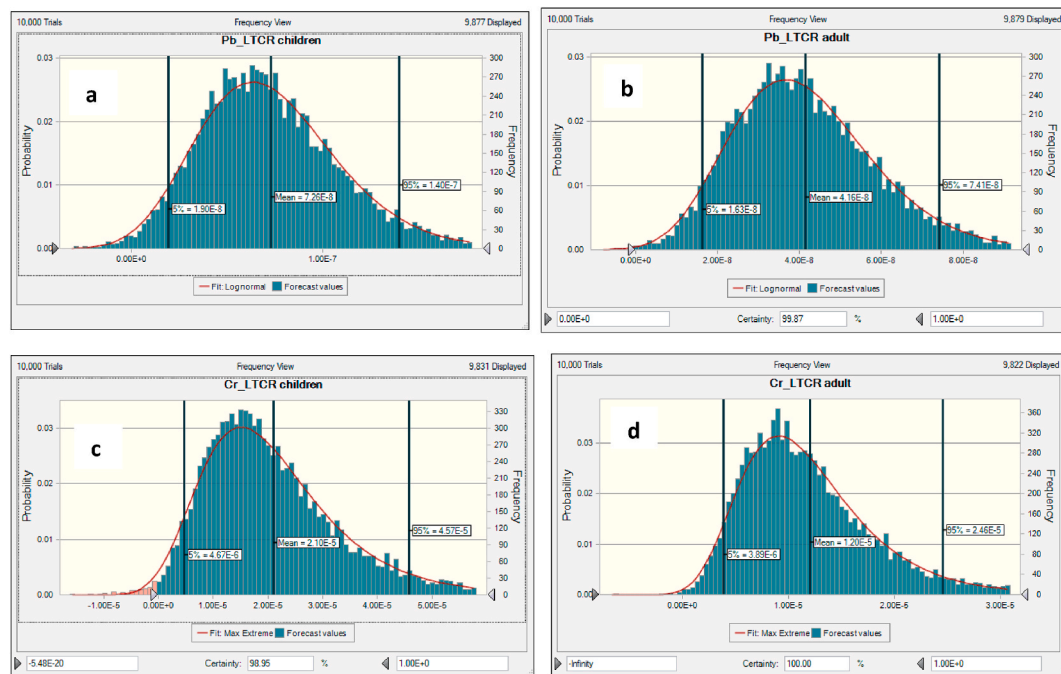


Fig. 7. Monte Carlo simulation for carcinogenic risk in BHs (a) exposure to Pb in case of children (b) and adults (c) exposure to Cr in case of children (d) and adults.

3.5. Sensitivity analysis

Sensitivity analysis is widely applied in risk management actions and decision-making. It was adopted in this study in order to ascertain the primary contributors to risk outcome. The sensitivity analysis showed that ingestion rate (IR) had the strongest positive effect on carcinogenic risk estimation by exposure to Pb and Cr in HWs (Figs. S5 and S6) and Pb and Cr in BHs (Figs. S7 and S8), respectively. However, the frequency (EF) of exposure and concentration of metal also had a significant impact on the risk outcomes in both cases.

4. Conclusion

The aim of this research is to look into the levels of heavy metals found in groundwater sources used in the study area, identify potential sources of heavy metals in groundwater, as well as estimate the health risk posed by exposure to such heavy metals using deterministic and probabilistic methods. Water samples were collected from 16 different locations which consist of 8 Hand dug wells (HWs) and 8 Boreholes (BHs) and analyzed for Heavy metals (HMs). The order of magnitude of the average contents for heavy metals are $Zn > Fe > Al > Cu > Cr > Mn > Pb$ in HWs and $Zn > Fe > Cu > Al > Cr > Pb > Mn$ for BHs. Pollution

indices reveal that BHs were of better quality while HWs were more polluted attributed to Pb, Zn and Cr. Source apportionment show presence of HMs are of anthropogenic and natural origin. The Monte Carlo Simulation (MCS) technique was used to assess the risk of non-carcinogenic and carcinogenic substances in children and adults. Upon exposure to chromium in the case of HWs, the respective mean and 95th percentile levels are 1.72 and 2.92 for children and 0.75 and 1.29 for adults indicating health risk (non-carcinogenic). Exposure to Cr (VI) has been linked to an increased risk of cancer in humans. The probability of exceeding this value is greater in children (about 90%) than in adults (16%). The prediction of potential lifetime cancer risk (LTCR) for Pb and Cr were also studied at both water source types. There is no carcinogenic risk associated with exposure to Pb because the calculated lifetime carcinogenic risk (LTCR) values via oral pathways are well below the 10^{-4} threshold. However, LTCR values for Cr exposure in children ranged from 0 to 2.14×10^{-4} for HWs indicating that exposure to present Cr levels poses a threat especially amongst children. Hence, drinking (ingestion) of these water sources for a prolonged duration increases the possibility of having cancer. Sensitivity analysis show that Exposure frequency and HMs concentration are the most significant input factors on the risk estimate. Overall, BHs represent a safe water source for both adults and children, while HWs could pose health risk in children due to elevated HMs levels especially Cr. Exposure to Cr has been associated with an increased risk of cancer in humans, particularly in the respiratory, urinary, bone, thyroid, and testicular systems. The use of probabilistic health risk modeling and spatial techniques to evaluate the most vulnerable population subject to risk (both carcinogenic and non-carcinogenic) should be further investigated.

Credit author statement

Ibitogbe Enoch: Conceptualization, Writing – original draft. Elemile Olugbenga O: Supervision and Methodology, Obayomi Kehinde Shola: Writing – review & editing Trzcinski A: Review and editing, Olajide: study design, Falana Michael: writing final manuscript, Okon Ekemini: Writing – review & editing, Arije Damilare: Writing – review & editing, Tokula Blessing: writing final manuscript, Ibitoye O: Writing – investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that might have had an impact on the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gsd.2022.100843>.

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